



Course Contents

EEE 321: Power System I

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Course Contents

Introduction to Power Systems

Elements of Power Systems, Power system representation, Bangladesh power system, Single Line Diagram, Impedance Diagram, Per unit system.

Transmission Line Parameters

Inductance calculations, single phase and three-phase system, GMD and GMR calculations, Capacitance calculations of single phase and three-phase lines.

Overhead Transmission Lines

Analysis of short, medium and long transmission lines, transmission line as a two-ports network, skin effect, Ferranti effect, corona effect, control of transmission systems, FACTS devices, Power factor improvement, HVDC. Sag and tension analysis, supports of different levels, insulators of overhead lines, potential distribution of overhead suspension insulator string.

Underground Cables:

Insulation resistance, capacitance, dielectric stress, grading of cables, capacitance of three phase cables, most economical conductor size.

Power Distribution System

Basic distribution systems and distribution voltage levels, distributor calculation of radial feeders, distribution substation and feeder, types and factors affecting distribution losses, different distribution networks.

Introduction to Switchgear and Protection

Circuit breaker, fuse, relay, overcurrent protection, unit protection and distance protection.

Introduction to Power Plant Terminologies

Introduction to power plants: types, operation, renewable integration challenges, and smart grid overview, Load duration curve, load factor, plant factor.

Introduction to Load Flow Analysis

Introduction to load flow/power flow, Bus Admittance Matrix, Gauss-Seidel Power Flow Solution

Infographic layout of course contents

Representation of Power System:

Network elements, Single line diagram of power system;

Parameters of Power System:

Inductance and capacitance of power transmission line;

Representation of Line:

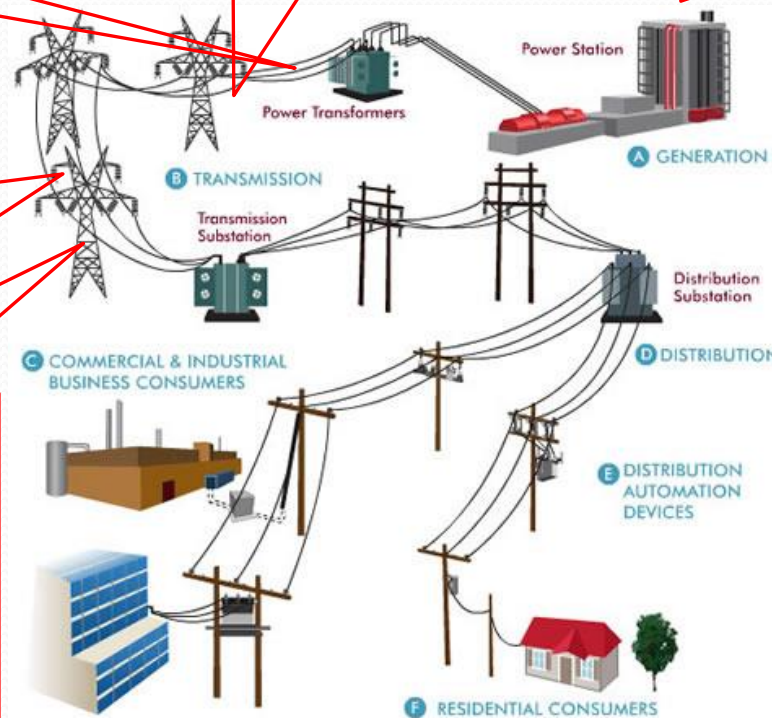
Equivalent circuit of short, medium and long lines, transmission line as a two ports network; skin effect; Ferranti effect; Corona effect; control of transmission systems, FACTS devices, HVDC.

Mechanical characteristic of overhead transmission line:

Sag and tension analysis; supports of different levels, insulators of overhead lines; Potential distribution of overhead suspension insulator string;

Power Plant Terminologies

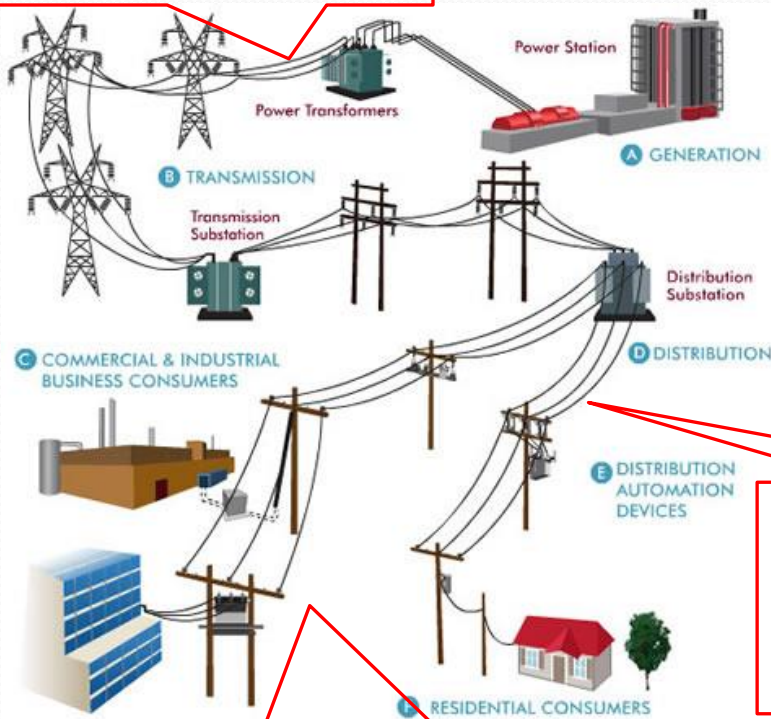
Load curve, load factor, plant factor



Infographic layout of course contents

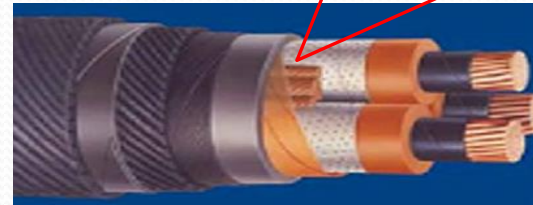
Load Flow Analysis

Bus Admittance Matrix, Gauss-Seidel power flow



Underground Cables:

Insulating resistance, capacitance, dielectric stress, grading of cables, capacitance of three phase cables, most economic conductor size.



Distribution of Power:

Basic distribution systems and distribution voltage levels, distributor calculation of radial feeders, different distribution networks.

Introduction to Switchgear and Protection

Circuit breaker, fuse, relay, overcurrent protection, unit protection and distance protection.

Course Outcomes

Upon successful completion of this course, students will be able to

Sl.	CO Description
CO1	Explain the fundamental concepts of power systems, including Bangladesh power system, per unit system, power plant operations with key terminologies, HVDC, FACTS devices, and power system protection principles.
CO2	Analyze transmission line parameters, calculate ABCD constants, solve transmission line and control problems, and perform load flow analysis using the Gauss-Seidel method.
CO3	Use Power system analysis tools to study transmission and distribution

Learning Materials

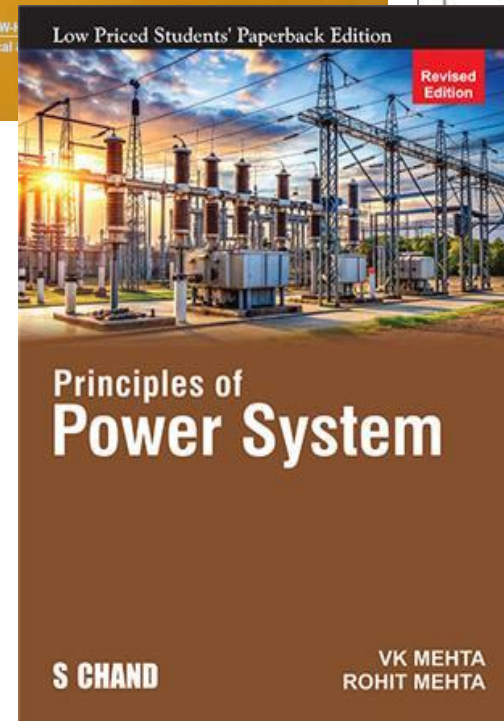
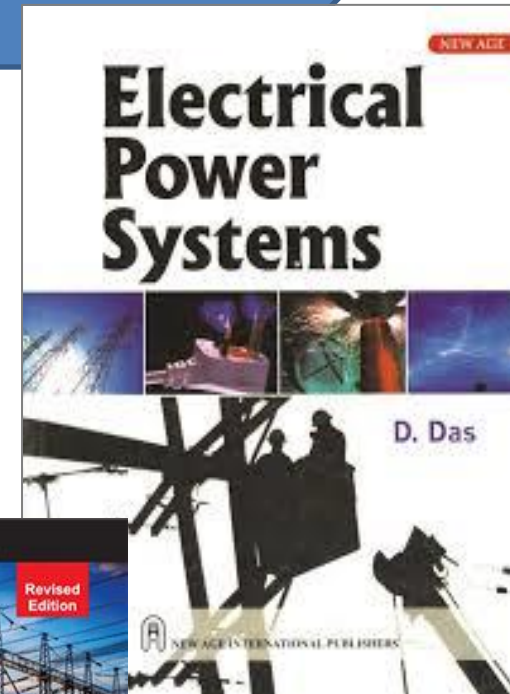
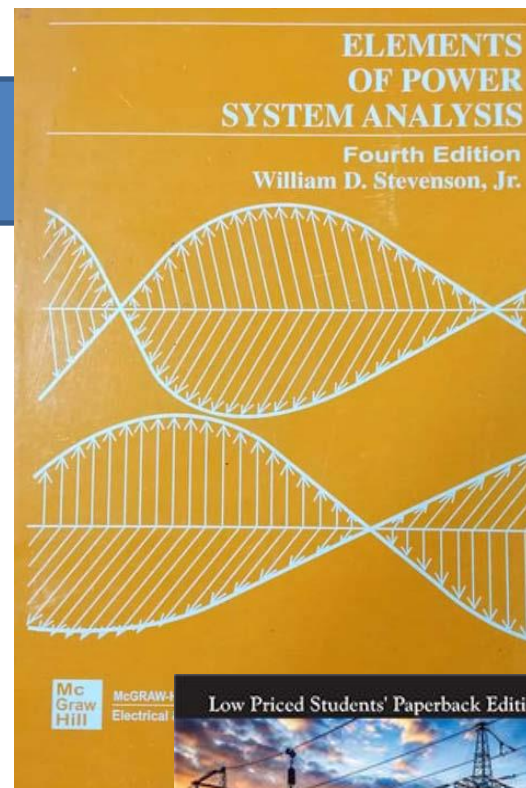
Course Text

Elements of Power System Analysis, W. D. Stevenson, 4th ed., McGraw Hill

Supplementary Readings

Principles of Power System, V.K. Mehta & Rohit Mehta, 4th ed., S. Chand Publishing, 2005.

Electric Power System by D. Das, New Age International Limited, 2006.



Assessment

CO Assessment Plan

Assessment Tools	CO1	CO2	CO3
Assignment	✓	✓	
Quiz	✓	✓	
Midterm Exam	✓	✓	
Final Exam	✓	✓	
Lab test, report, project			✓

Marks Distribution

Assessment Tools	Weightage (%)
Class Performance & Attendance (70%)	10
Quizzes (Best 2 out of 3)	20
Midterm Exam	20
Final Exam	30
Assignment/Project	20
Total	100

Course Teacher



Inspiring Excellence

Dr. Tareq Aziz

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Research Interests:

Power System Stability, Renewable Integration into grid, Demand Side Management, Electric Vehicle, Signal Processing.

Google Scholar:

<https://scholar.google.com/citations?user=FN628KAAAAAJ&hl=en>



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